

Counter Strike 2 Event Modelling Using Mutually Exciting Point Processes

A project report
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1 Introduction

1.1 Description MEPP Model

A mutually exciting point process, more commonly known as a Hawkes process, is primarily used to model situations where events are likely to trigger subsequent events, creating clustering patterns [3]. This indicates that the Hawkes process would be applicable to time-dependent events. To further prove this, the Hawkes process has previously been shown to model sports markets, attempting to determine winners of a basketball game based on a variety of factors which fit into the model, including, but not limited to: home / away team, the "hot shooter" effect, and rivalries [2].

The intensity function is a fundamental concept in the study of point processes. Given the history of the process up to that point in time, the function characterizes the instantaneous rate at which events are expected to occur. By tinkering with the model parameters (Section 2.2) one can model different types of event clustering behavior.

In summation, the intensity function provides a flexible framework for modeling complex temporal dependencies in event data, making it a powerful tool for applications where understanding the dynamics of event occurrences is crucial. This function will be the primary application of research in this paper.

1.2 Description of Conducted Research

A white paper written by Xin-Yu Tian and Jian Shi [2] was studied to get an informative, applied understand of a MEPP model. Tian & Shi model basketball games using marked mutually exciting point processes. The research in this paper was adapted from Tian and Shi to the e-sport *Counter-Strike 2* (CS2). The transition from traditional sports to e-sports presents potential opportunities, and CS2, with its structured game-play and dynamic interactions, serves as a candidate for applying the MEPP model. The game's inherent characteristics, such as frequent player interactions and event-driven dynamics, align well with the self-exciting nature of the Hawkes process, making it a suitable framework for modeling and predicting in-game events. The most frequent of all game events has been chosen to be modeled: elimination-sequences. See the section below for more information on the rule specifics of the game. These elimination-sequences can happen in very small time frames, making events "excitable" in nature.

1.3 Rules of Counter Strike 2

For context to the reader, this is a description of the rules of CS2. If the reader is familiar, they should skip this section. The basic rules of CS2 are:

- 5v5 team-based shooter
 - Team 1 is called Terrorists (T)
 - Team 2 is called Counter-Terrorists (CT)
- Each game is divided into 26 potential rounds
- Each game is divided into two halves, switching at the end of round 13
 - E.g., Team A starts as T, switches to CT at end of round 13
- First team to win 13 rounds wins the game
- An individual round lasts 1 minute 55 seconds
- A round can be won by:
 - Eliminating the other team before time runs out
 - Planting the bomb (T), Defusing the bomb (CT)

Elimination sequences are a core mechanic of the game. In this research, an elimination sequence is defined as the amount of eliminations which occur in **two seconds**.

An example of an elimination sequence could be as follows:

1. Player 1 Eliminates Player 2
2. Player 3 Eliminates Player 1
3. Player 4 Eliminates Player 3

All of which happened in 0.5 seconds.

The game encourages strategic depth by offering a wide variety of maps. The combination of time-constrained rounds, team dynamics, diverse environments, and the "hot shooter" effect create a tapestry of events that are both similar to basketball and can be modeled using the MEPP framework.

2 Mathematical Derivations

2.1 Hakes Process Definition

A Hawkes process is defined as a point process $N(t)$ with intensity function $\lambda(t)$ that takes the following form:

$$\lambda(t) = \mu + \sum_{t_i < t} g(t - t_i)$$

where:

- $\lambda(t)$ is the **conditional intensity function** at time t
- $\mu > 0$ is the **baseline intensity** (background rate)
- t_i are the **timestamps of previous events**
- $g(t)$ is the **decay kernel** (excitation function)

The most commonly used decay kernel is the exponential function:

$$g(t) = \alpha e^{-\beta t}$$

where:

- α controls the **jump size of intensity** at each event
- β determines the **decay rate of the excitement**

Thus, the complete intensity function is given by:

$$\lambda(t) = \mu + \alpha \sum_{t_i < t} e^{-\beta(t-t_i)}$$

To calculate the probability of an event occurring in a small interval $[t, t + \Delta t]$, the intensity function $\lambda(t)$ is used. The probability of observing an event in this interval is approximately:

$$\mathbb{P}(\text{event in } [t, t + \Delta t]) \approx \lambda(t)\Delta t$$

This approximation is valid for sufficiently small Δt , reflecting the instantaneous rate of event occurrence as dictated by the intensity function.

2.2 Parameter Effects

Each parameter in the model influences how events are predicted to occur over time:

The **baseline intensity** μ acts like a constant background noise level. It represents the average rate at which events happen on their own, without being triggered by previous events. A higher μ means that events are more likely to occur randomly, even if nothing has happened recently.

The **jump size** α is a measure of how much excitement or influence each event has on future events. When an event occurs, α determines how much it increases the likelihood of more events occurring soon after. A larger α means that each event has a stronger impact, leading to more clustering of events.

The **decay rate** β is similar to a timer that counts down the influence of past events. It controls how quickly the excitement of a past event fades away. A larger β means that the influence of an event disappears quickly, while a smaller β allows the effect to linger, keeping the intensity elevated for a more extended period.

2.3 Properties of Hawkes Process

- **Stationary condition:** $\frac{\alpha}{\beta} < 1$
 - This condition ensures the process remains stable and doesn't explode to infinity
 - When this condition is met, the number of events stays manageable over time

- **Expected intensity:** $\mathbb{E}[\lambda(t)] = \frac{\mu}{1-\alpha/\beta}$
 - This represents the average rate at which events occur in the long run
 - The formula shows how the baseline rate μ is amplified by the self-exciting effects
- **Branching ratio:** $\eta = \frac{\alpha}{\beta} = \int_0^\infty g(s)ds$
 - This represents the average number of "daughter" events triggered by each event
 - A branching ratio of 0.5 means each event triggers, on average, 0.5 new events

3 Implementation

This implementation of MEPP was used to model elimination sequences, with the purpose of obtaining the values of the intensity function and correlating those to conditional probabilities of the next elimination occurring over the next X seconds. For the purpose of this project, a model was created for an individual team for an individual map. This was chosen because of the nature of the game. Teams have various play styles on different maps. Thus, the professional team Natus Vincere, on arguably the most famous first-person shooter map of all time, *Dust II* [1], were used to produce a model.

3.1 Description of Data

The replay files of ten professional matches were harvested from <https://hltv.org>. The game publisher, Valve, has made custom *.dem* files of every historical professional match containing all the metadata about what occurred in a game. For the concerns of this project, a parser was made in python which extracted the following information:

- Elimination event (E.g., Player A eliminated player B)
- Which side the elimination happened (E.g., T eliminated a CT)
- Location of elimination (E.g., (10.5, 11.2, 114.6))
- Distance of elimination (E.g., Players were 400 units away)

- Time, in seconds, when the elimination occurred, relative to when the match started (E.g., 4.2 seconds since match start)

Full Example: Team Natus Vincere, playing the attacker side in (x,y,z) location, eliminated player ABC in (x,y,z) location from 245 units away at 50 seconds after game start.

3.2 Team and Spatial Components

While Section 2.1 explains the base derivation of the model, the implementation was adapted for a couple of key characteristics which are centric to Counter-Strike 2. As mentioned in Section 1.3, it's crucial to understand that each team plays two halves in one single game. This naturally encourages different play styles depending on which team is playing on what side.

The other considered factor was the distance at which engagements are taken. Taking a close range engagement vs an extremely long range engagement should be considered as a factor when computing the intensity of a given event, as distance effects the probability of a given elimination sequence.

There are many other situations which one could experiment with which have an effect on the intensity, however, for the purpose of this research, only the aforementioned scenarios were accounted for.

3.3 Modified Hawkes Process

For a given point in space-time (t, x, y, z) , the team-specific intensity functions is defined as follows:

$$\lambda_T(t, x, y, z) = \mu_T + \sum_{(t_i, x_i, y_i, z_i) \in H_T, t_i < t} \alpha_T \cdot e^{-\beta(t-t_i)} \cdot S(x, y, z, x_i, y_i, z_i) \quad (1)$$

$$\lambda_{CT}(t, x, y, z) = \mu_{CT} + \sum_{(t_i, x_i, y_i, z_i) \in H_{CT}, t_i < t} \alpha_{CT} \cdot e^{-\beta(t-t_i)} \cdot S(x, y, z, x_i, y_i, z_i) \quad (2)$$

where:

- $\lambda_T(t, x, y, z)$ and $\lambda_{CT}(t, x, y, z)$ are the conditional intensity functions for the Terrorist and Counter-Terrorist sides, respectively, at time t and position (x, y, z)
- $\mu_T > 0$ and $\mu_{CT} > 0$ are the baseline intensities for each side
- α_T and α_{CT} are the excitation parameters for each side
- β is the temporal decay rate (common to both sides)
- H_T and H_{CT} represent the history of events for each side
- (t_i, x_i, y_i, z_i) denotes the time and spatial coordinates of a past event

The spatial decay function $S(x, y, z, x_i, y_i, z_i)$ captures how the influence of past events diminishes with physical distance:

$$S(x, y, z, x_i, y_i, z_i) = e^{-d((x, y, z), (x_i, y_i, z_i))/\sigma} \quad (3)$$

$$\text{where } d((x, y, z), (x_i, y_i, z_i)) = \sqrt{(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2} \quad (4)$$

Here:

- $d((x, y, z), (x_i, y_i, z_i))$ is the Euclidean distance between the current position (x, y, z) and the position of the past event (x_i, y_i, z_i)

- σ is the spatial scale parameter, which controls how quickly the influence decays with distance

This extended model captures several important aspects of the game dynamics:

1. **Team-specific parameters:** The baseline intensities (μ_T, μ_{CT}) and excitation parameters (α_T, α_{CT}) allow for asymmetric behavior between teams.
2. **Temporal clustering:** The exponential decay term $e^{-\beta(t-t_i)}$ models how the influence of past events diminishes over time.
3. **Spatial clustering:** The function $S(x, y, z, x_i, y_i, z_i)$ captures how events are more likely to occur near locations of past events, reflecting the tactical nature of positioning in Counter-Strike 2.

The probability of observing an event from team $j \in \{T, CT\}$ in a small space-time volume $[t, t + \Delta t] \times [x, x + \Delta x] \times [y, y + \Delta y] \times [z, z + \Delta z]$ is approximately:

$$\mathbb{P}(\text{event from team } j \text{ in small volume}) \approx \lambda_j(t, x, y, z) \cdot \Delta t \cdot \Delta x \cdot \Delta y \cdot \Delta z \quad (5)$$

This model provides a framework for understanding and predicting the spatio-temporal dynamics of events in Counter-Strike 2 matches.

3.4 Backtesting Framework

A back-testing framework was created using python allowing the integration between parsing historical data, calculating the intensity functions & associated probabilities, and plotting the results. Please see the diagram below for software specifics.

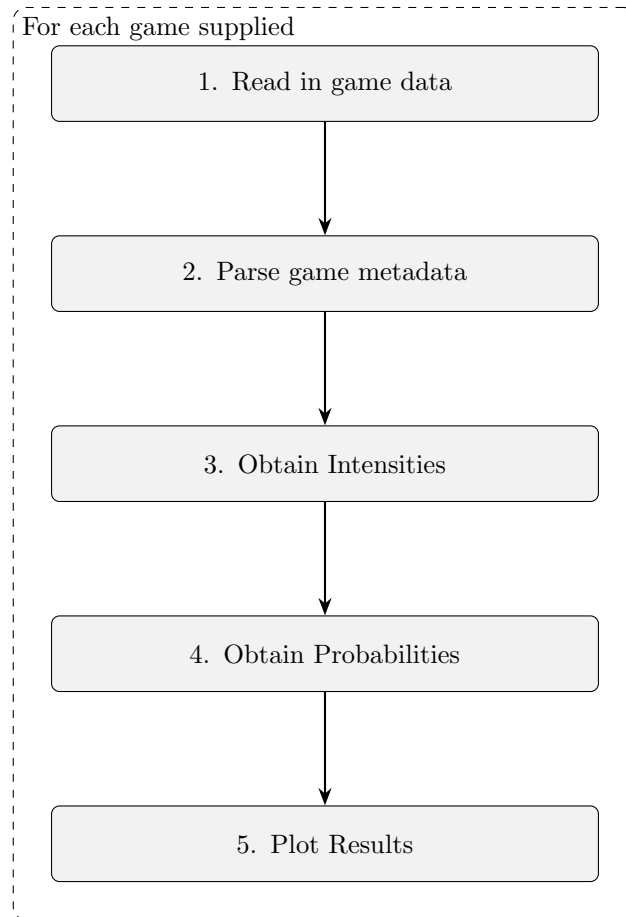


Figure 1: *Software architecture for game analysis using the Hawkes process model*

3.5 MonteCarlo Simulations

Lastly, a MonteCarlo simulation framework was created such that the various parameter values could be tested to obtain the most accurate model. Please reference the diagram below to see the general architecture of the simulator.

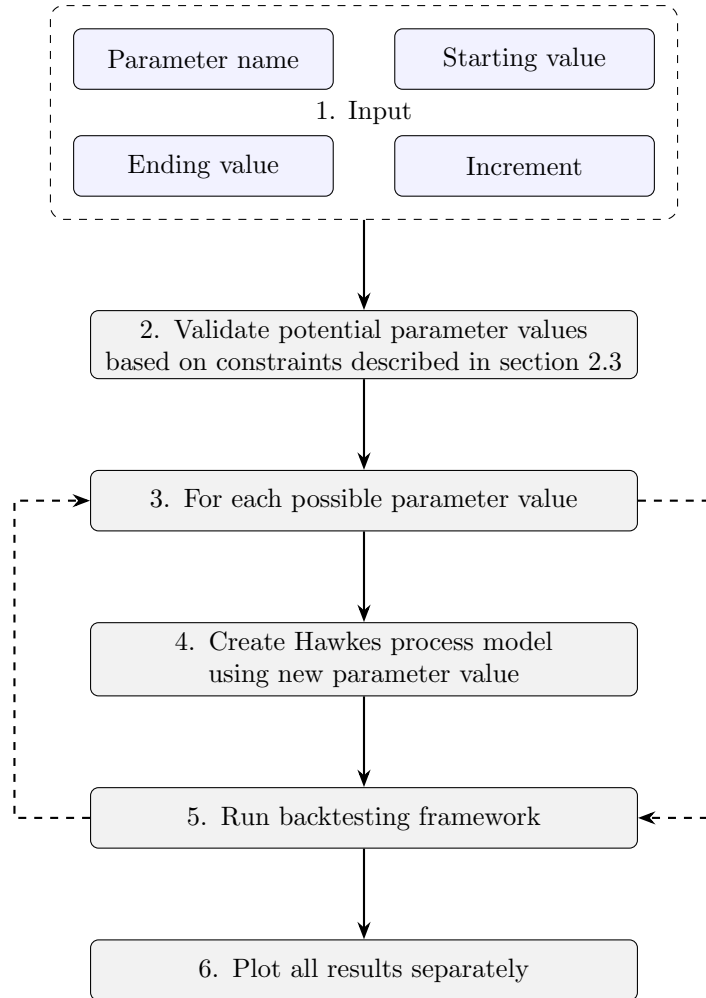


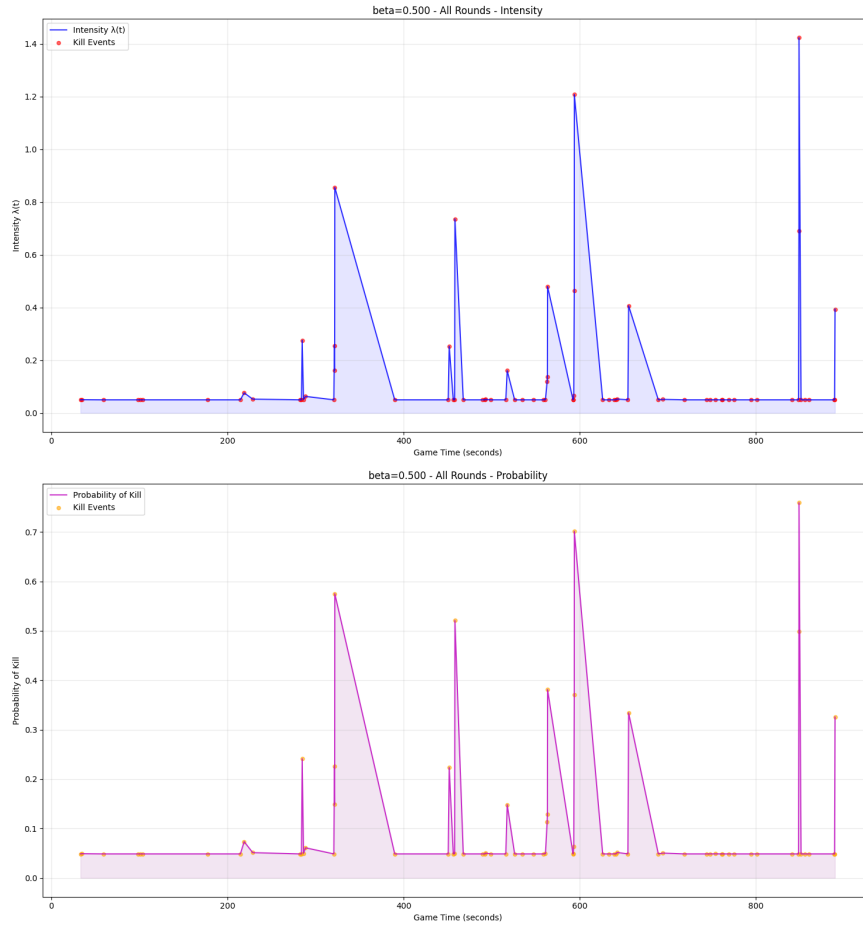
Figure 2: *Parameter optimization workflow for the Hawkes process model*

4 Results

The results in this section are shown using the following parameters (see table below), which were the output from running monte carlo tests. For brevity, I have presented the graphs of a single game and select rounds which clearly highlight the behavior of the intensity and probability.

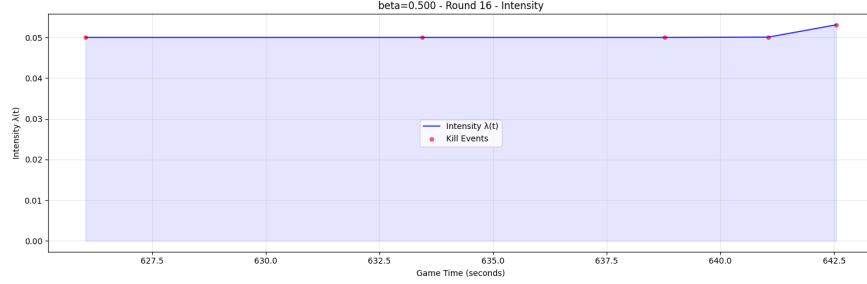
μ_T	μ_{CT}	α_T	α_{CT}	β	spatial_scale
0.05	0.05	0.8	0.8	0.5	100

Table 1: *Optimal values used for the model*

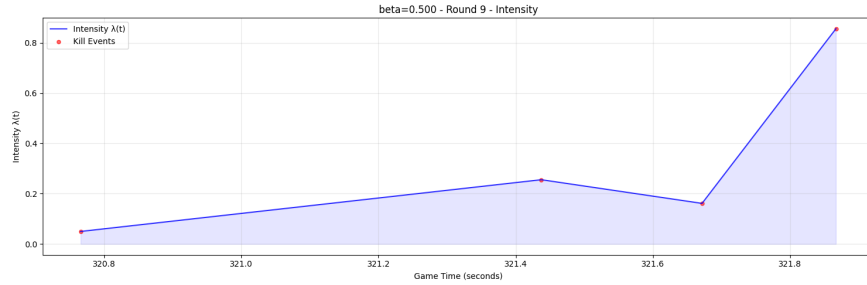


The above results show that, as the events occur in quick succession, the intensity of the kernel function increases dramatically. While the model is providing legitimate results, viewing a full game at a time makes it hard to determine what is going on during rounds where elimination events are not done in quick succession.

To clearly understand what is occurring, let's compare two rounds' intensity functions. The first will be a round where many eliminations happened in quick succession; the second will show a slower rate of eliminations.



(a) Intensity for a round with sporadic events



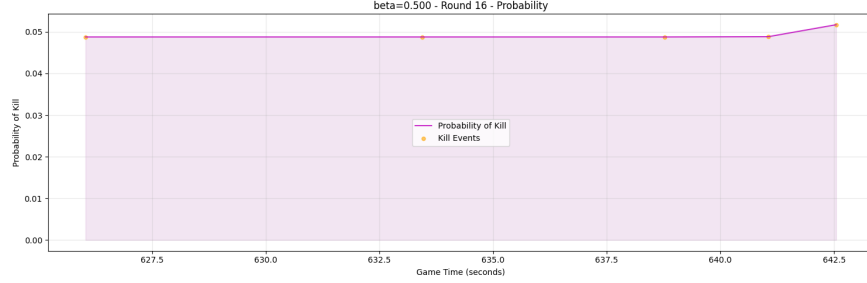
(b) Intensity for a round with events in quick succession

What's important to notice here is the scale of both the X and Y axis. In a round with low intensity, the results remain very close to our baseline intensity, or $\mu_t = \mu_{ct} = 0.5$. This indicates that the game is being played to a high level of skill, and that, even though the team gets an elimination event, that the enemy is playing well and responding appropriately.

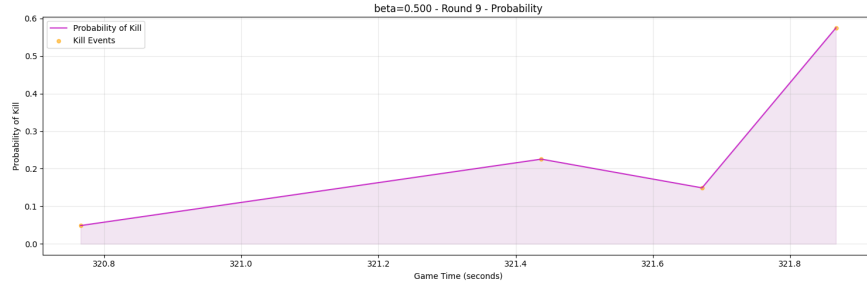
However, high intensity rounds occur when eliminations happen in sub second intervals. This reflects the natural progression of the game. The mechanic involved is called "trading." Where, for a given team, if one of your teammates gets eliminated, you try and eliminate the person who eliminated your teammate, so on and so forth. It is a domino-like effect. The skill comes into play when one team defies the natural progression of a "trade." Thus achieving multiple eliminations in a short amount of time, greatly increasing the chances for a round win.

The key to high intensity rounds is the notion to defy trading elimination events. Meaning, the team has the "hot shooter" effect. In MEPPs terminology, this is reflected by having a higher valued intensity function at that point in time.

Below is the probability readings for the same rounds:



(a) Probabilities for a round with sporadic events



(b) Probabilities for a round with events in quick succession

This correlates to our intensity readings, with the highest probability of another event maxing out around 55%, and the lowest being close to 4%. Although many other games and rounds were analyzed, they all look pretty much similar to this.

Looking forward, a promising direction for this model would be the implementation of a real-time betting model based on the probabilities of elimination sequences. Such an advancement would necessitate establishing correlations between elimination patterns and round outcomes, and subsequently, determining how these round results influence the probability of match victory. While much more development would be involved, it is possible due to the nature of live, in-game betting which is allowed in online sports-books.

5 Conclusion

This application of MEPP to CS2 represents an extension of stochastic modeling into the realm of esports, an area previously unexplored with this mathematical framework. This endeavor significantly stretched my mathematical capabilities, requiring adaptation of concepts to the unique dynamics of CS2. While the results demonstrate the model’s potential, they also reveal areas for refinement, particularly regarding the temporal decay function. The observed patterns suggest that the exponential distribution may not fully capture the complex temporal dependencies present in CS2.

The process of modifying the MEPP to incorporate team-specific parameters and spatial dimensions proved to be challenging yet rewarding, substantially increasing my understanding of stochastic processes and their applications. Developing a Monte Carlo simulation and a backtesting framework provided experience with cutting-edge modeling techniques that are broadly applicable across various statistical modeling domains.

In short, this was a fascinating topic and extremely rewarding to learn and implement.

References

- [1] TheGamer. *First-Person Shooter Best Multiplayer Maps*. Online. URL: <https://www.thegamer.com/first-person-shooter-best-multiplayer-maps/>.
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- [3] Wikipedia contributors. *Hawkes process*. Wikipedia, The Free Encyclopedia. Accessed: 2023-11-09. 2023. URL: https://en.wikipedia.org/wiki/Hawkes_process.